

LARGE HYDROELECTRIC STORAGE FACILITIES

DRAFT — DBA-ES-HY-FC1: Large Hydroelectric Storage Facilities

*Design Basis, Design Basis Accident, and Military Action
Extension*

Version 0.5 — DRAFT

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Tim Roxey

Eclectic Technologies

*For review by GenCo engineers, FERC dam safety staff, NERC reliability analysts, and policy officials
responsible for electric sector critical infrastructure protection.*

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Position in hierarchy: Facility-class instrument. All sector-level methodology is inherited from DBA-ES. This document applies that methodology to large hydroelectric storage facilities and adds dual-consequence-chain analysis specific to this facility class.

Section 1 — Purpose, Scope, and Inheritance

1.1 Purpose

This document establishes the Design Basis for large hydroelectric storage facilities within the Electric Sector Design Basis Series. It is the first facility-class instrument produced under the DBA-ES sector parent (v0.9) and serves as the methodology template for subsequent facility-class instruments in the series.

The document applies the DB → DBA → DBA-MA three-layer analytical progression to large hydroelectric storage facilities: a baseline Design Basis (DB) that establishes postulated events from equipment failure, natural hazards, and operational conditions; a Design Basis Accident (DBA) layer that bounds the non-adversarial consequence envelope and establishes acceptance criteria; and a military action extension (DBA-MA) that applies the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to each postulated event category.

Large hydroelectric storage facilities present a consequence structure not encountered in other electric-sector facility classes: a dual-consequence chain in which a single initiating event produces simultaneous consequences in two distinct domains. The generating consequence — loss of a large GenCo asset whose failure is a bulk electric system reliability event — propagates through the electric grid. The dam safety consequence — potential failure of the impoundment structure, resulting in a downstream flood wave — propagates through the watershed's physical geography. Under military action conditions, both chains can be initiated simultaneously by a single precision strike or coordinated attack. This dual-chain structure is the central analytical contribution of this facility-class instrument and has no analog in the sector-level DBA-ES parent document.

1.2 Facility Class Definition

Large hydroelectric storage facilities within the scope of this instrument are defined by three criteria, all of which must be satisfied:

- Installed generating capacity exceeds 500 megawatts (MW) connected to the bulk electric system at transmission voltage.
- The facility incorporates an impoundment structure — a dam — that retains a storage reservoir of sufficient volume that dam failure would produce a downstream flood wave with significant life safety and infrastructure consequences. Operationally, this corresponds to the FERC dam safety High Hazard Potential classification or an equivalent under the applicable state dam safety program.
- The generating units are conventional synchronous machine turbine-generators that contribute synchronous inertia to the interconnection. This criterion excludes pumped storage facilities operating in pump mode, which contribute no synchronous inertia during pumping operations and present a different consequence profile.

Reference facilities that satisfy these criteria include Grand Coulee (Washington, 6,809 MW), Robert Moses Niagara (New York, 2,675 MW), Hoover (Nevada/Arizona, 2,080 MW), Glen Canyon (Arizona, 1,320 MW), John Day (Oregon/Washington, 2,160 MW), and Chief Joseph (Washington, 2,620 MW), among others. These facilities collectively represent a significant fraction of the synchronous inertia base in the Western Interconnection and contribute substantially to frequency response in both the Western and Eastern Interconnections.

1.3 Run-of-River Facilities

Run-of-river hydroelectric facilities — those that divert water flow for generation without maintaining a significant storage reservoir — are outside the scope of this instrument. Their consequence profile differs from that of storage facilities in two material respects: the absence of a significant impoundment means that facility loss does not present a dam-failure flood consequence, and the absence of reservoir storage means that flow regulation for downstream water supply, navigation, and irrigation is absent.

Run-of-river facilities whose generating capacity meets the 500 MW threshold are subject to the DBA-ES sector-level framework and the MA extension applicability triggers established in DBA-ES Section 1.5. Where those triggers are met, the methodology in this instrument applies with the dam safety consequence chain set to null. A companion instrument or annex addressing run-of-river facilities above 500 MW is anticipated as the series matures. For the current revision, run-of-river facilities are DBA-complete under the DBA-ES sector framework alone.

1.4 MA Extension Applicability

Per DBA-ES-MA-SF v0.2 §1.4 and DBA-ES-MA-QB v0.1 §5, each DBA-ES FC instrument must present a structured four-filter applicability assessment before the operator engages with the MA extensions. For DBA-ES-HY-FC1, the assessment is resolved at the class level as follows.

- **Filter 1 — Geolocation relative to defense-critical loads: SATISFIED BY DEFINITION. Large hydroelectric storage facilities with capacity ≥ 500 MW serve**

broad regional load areas that include defense-critical facilities, military installations, and continuity-of-government sites as a matter of geography. Grand Coulee, Hoover, Robert Moses Niagara, and comparable facilities supply entire metropolitan regions, including federal facilities. The downstream flood wave from dam failure (Section 7.3) may directly inundate defense-critical facilities in the flood zone regardless of electrical consequences.

- Filter 2 — Position in strategic transmission corridors: SATISFIED BY DEFINITION. Facilities at the ≥ 500 MW threshold connect to the bulk electric system at EHV voltage through switchyards that are themselves nodes in strategic transmission corridors. The generating switchyard at a large hydro facility is typically a CIP-014 Tier 1 or Tier 2 asset. The loss of the generation block simultaneously removes a large transmission-connected generation source and may require reconfiguration of the transmission topology in the affected corridor.
- Filter 3 — Substitutability: VERY LOW. Unit step-up transformers at large hydro facilities are custom EHV equipment with RT_HVLLC of 18 to 30 months. Turbine runners and generator stator windings for large units have comparable lead times. The dam structure itself — if breached — has no replacement pathway; dam reconstruction following catastrophic failure is a multi-year civil engineering project with no precedent in the U.S. for a facility of this scale. The dam failure consequence chain (Section 7.3) has zero substitutability: there is no equivalent downstream flood wave mitigation measure once the dam has failed.
- Filter 4 — Adversary targeting logic: HIGHEST CONSEQUENCE PHYSICAL TARGET IN THE DBA-ES SERIES. No other DBA-ES facility class produces a simultaneous generating consequence and a catastrophic physical geography consequence from a single precision attack. The dam breach consequence chain — downstream flood wave, reservoir drainage, watershed contamination, loss of cooling water for downstream nuclear facilities — represents a consequence magnitude with no analog in any other DBA-ES instrument. The Kakhovka Dam destruction (June 6, 2023) confirmed that this facility class is an active military target and that the dual-consequence attack is operationally achievable by a near-peer adversary.

Conclusion: The MA design basis applies to all DBA-ES-HY-FC1 facilities. No facility-level exemption assessment is required. The MA extension in Section 6 is a mandatory component of this instrument for all facilities meeting the Section 1.2 criteria.

1.4a MA Communications Design Basis

DBA-ES-MA-QB v0.1 §5 establishes the MA communications design basis as a sector-level requirement applicable to all DBA-ES FC instruments. The MA communications baseline condition — primary SCADA and voice communications channels are unavailable — is the assumed design basis condition for all MA scenario analysis in this instrument. The following

instantiates CB-1 through CB-5 for the HY-FC1 facility class, with a note specific to the remote siting character of most large hydroelectric storage facilities.

FC1-specific communications note: Large hydroelectric storage facilities are typically remotely sited with limited communications infrastructure redundancy. Under MA conditions, the combination of physical attack and communications interdiction can leave facility operators without contact with the CntlCo, the dam safety authority, and emergency services simultaneously. The CB-2 local authority requirement is therefore the most critical CB obligation for FC1: the design basis must ensure that dam safety emergency actions — EAP activation, spillway gate operation, emergency reservoir drawdown — are executable by on-site operators without external communications, because communications loss and physical attack may be simultaneous.

CB-1 (Backup channel): The facility must maintain at least one backup communications channel to both the serving CntlCo and the dam safety authority (FERC, USACE, or USBR as applicable) that is architecturally independent of primary SCADA and voice infrastructure. The two channels serve distinct obligations — CntlCo contact for grid emergency coordination, dam safety authority contact for EAP activation — and must both satisfy CB-1 independently. Acceptable channels: dedicated licensed radio, satellite communications, or physically diverse fiber.

CB-2 (Local authority): On-site operators must have pre-authorized authority to execute all dam safety emergency response procedures — including EAP activation, spillway gate operation, powerhouse shutdown, and reservoir emergency drawdown — without CntlCo direction, dam safety authority contact, or any external communications. This is the most operationally significant CB obligation for FC1: under MA conditions, the dam safety consequence chain cannot wait for communications restoration. The pre-authorized local authority framework must be documented, exercised annually, and consistent with the FERC-approved Emergency Action Plan.

CB-3 (Command Broker continuity): MA communications degradation does not suspend the Command Broker obligation for above-Bright-Line Gen AI advisory tools. The qualified hydro facility operator on duty remains the mandatory Command Broker. Above-Bright-Line Gen AI recommendations for gate operations, generation dispatch, or reservoir management must not be actioned without licensed operator confirmation, regardless of communications status.

CB-4 (Verification cadence): Both the CntlCo-facing and dam-safety-authority-facing backup channels must be tested at an interval specified in the facility's design basis implementation. The test condition must simulate the MA baseline: primary SCADA and voice communications unavailable at the facility simultaneously.

CB-5 (Degraded-mode procedures): FC1 emergency operating procedures must include a degraded-communications mode executable by on-site operators with no external contact. Procedures must address: (a) autonomous facility actions when all communications are lost; (b)

backup channel activation sequence; (c) dam safety actions executable without external authorization (consistent with CB-2); (d) threshold for initiating EAP notification through alternate means (runners, local sirens, coordination with local emergency management) when all electronic communications are unavailable.

1.5 Inheritance Map

The following table identifies which elements of this instrument are inherited from DBA-ES unchanged, which are inherited with facility-class adaptation, and which are locally defined. This map must be reviewed and updated at each revision of this instrument and of the parent DBA-ES document.

Element	Status	Notes
DB → DBA → DBA-MA three-layer progression	Inherited unchanged	Section 1 of this document
Four postulated event categories (PE, SE, CE, XC)	Inherited — instantiated	Section 4: hydro-specific events within each category
Three-tier consequence taxonomy	Inherited — extended	Section 7: dual-chain analysis adds downstream flood as a parallel Tier 1–2 consequence
MA extension applicability triggers.	Inherited unchanged	DBA-ES-HY-FC1 facilities meet Triggers 1, 2, and 3 by definition
Bright Line/command broker requirement	Inherited unchanged	Section 8: applied to the governor, excitation, and gate control systems
Recovery sequencing governing principle	Inherited unchanged	Section 8: CntlCo-coordinated black start sequencing added
GenCo/TransCo/DisCo/CntlCo terminology	Inherited unchanged	All hydro facilities in scope are GenCo assets
Consequence-chain organizing principle	Inherited unchanged	Primary chain: GenCo generation loss; secondary chain: dam failure flood wave
Acceptance criteria framework structure	Locally defined	Section 9: Physical, operational, and cross-sector criteria for hydro class
Gap analysis framework structure	Locally defined	Section 10: hydro-specific gap categories and investment prioritization logic
Dual consequence chain analysis	Facility-class specific	New element: no sector-level analog; addresses simultaneous GenCo and dam safety consequences under MA conditions.

Section 2 — Regulatory and Institutional Context

2.1 FERC Jurisdiction

The Federal Energy Regulatory Commission licenses large hydroelectric storage facilities operating in the United States under the Federal Power Act. FERC licenses establish operating conditions, minimum flow requirements, recreation obligations, and safety program requirements. The FERC dam safety program, administered under 18 CFR Parts 12 and 6, requires licensees to maintain current dam safety inspections, Emergency Action Plans (EAPs), and surveillance and monitoring programs. FERC Part 12 periodic inspections, conducted by independent Consultant Engineers on a five-year cycle, provide the regulatory mechanism for assessing and documenting dam safety compliance.

For DBA-ES-HY-FC1 purposes, FERC Part 12 provides the baseline regulatory framework for the dam safety consequence chain. The design basis analysis in this instrument extends FERC's safety program by adding postulated events driven by military action that FERC Part 12 does not currently address. This extension does not displace FERC's regulatory authority; it supplements the safety analysis with the adversarial threat tier that the existing program does not cover.

2.2 NERC Reliability Standards

Large hydroelectric storage facilities are registered with NERC as Generator Owners (GO) and Generator Operators (GOP) and are subject to NERC reliability standards applicable to those functional entity registrations. Relevant standards include:

- FAC-001 and FAC-002 (Facility Ratings): Establish the thermal and stability ratings of generating units and associated transmission facilities. The design basis for the generating consequence chain must be consistent with the facility's FAC-002 ratings.
- CIP-014 (Physical Security): Applies to transmission owners and operators, and by functional relationship to generators whose direct connection to the transmission system means their switchyard is within scope. The physical security plan required by CIP-014 R3 is a baseline input to the physical hardening gap analysis in Section 10.
- PRC-024 (Generator Frequency and Voltage Protective Relay Settings): Establishes the ride-through requirements for generating units under frequency and voltage disturbances. Under military action conditions that cause grid frequency disturbances, PRC-024 compliance determines whether units trip or ride through the disturbance. The relationship between PRC-024 and the DBA-MA frequency response scenario is addressed in Section 6.
- BAL-003 (Frequency Response and Frequency Bias Setting): Establishes the frequency response obligation for Balancing Authorities, which in turn determines the consequence assessment for the loss of a large synchronous GenCo asset. The BAL-003 framework is the quantitative basis for Tier 2 consequence assessment in Section 7.

2.3 Army Corps of Engineers and Bureau of Reclamation

Several large hydroelectric storage facilities within the scope of this instrument are owned and operated by federal agencies rather than private licensees. Grand Coulee and Chief Joseph are operated by the Bonneville Power Administration (BPA); Hoover and Glen Canyon are operated by the Bureau of Reclamation (USBR); and numerous facilities are owned by the Army Corps of Engineers (USACE). Federal ownership does not exempt these facilities from FERC dam safety oversight in all cases, but it does create a different regulatory relationship than applies to private licensees.

For DBA-ES-HY-FC1 purposes, federally owned and operated facilities are within scope. The design basis methodology applies without modification. The regulatory enforcement pathway for gap closure differs — it runs through agency capital planning processes and congressional appropriations rather than FERC license conditions — but the analytical framework is identical. The distinction is noted where it affects the gap analysis framework in Section 10.

2.4 USCOLD and Dam Safety Engineering Standards

The United States Society on Dams (USCOLD, operating as USSD — United States Society on Dams since 1999) publishes technical guidelines for dam safety engineering that FERC, USACE, and USBR reference in their dam safety programs. Relevant guidelines include the Recommended Guidelines for the Safety Inspection of Dams and the guidelines for Dam Safety Emergency Action Planning. These guidelines provide the technical standard against which the dam safety consequence chain analysis in Section 7 is assessed.

The design basis for dam safety under military action conditions extends beyond USCOLD guidelines, which address failure modes arising from equipment failure, flood, earthquake, and internal erosion, but do not address deliberate targeted destruction. Section 6 develops the military action extension for dam safety, using the same four-characteristic framework applied to the generating-consequence chain.

2.5 Institutional Context: DBA-ES-HY-FC1 and the Dams Sector

As established in DBA-ES Section 1.4, DBA-ES-HY-FC1 is positioned within the Electric Sector series based on the generating consequence chain as the primary consequence domain. This positioning does not displace the Dams Sector’s regulatory relationships — FERC, USACE, USBR, and state dam safety agencies remain the governing authorities for dam safety obligations. The transmittal message for this instrument to Dams Sector agencies is that the DBA-ES-HY-FC1 analysis complements existing dam safety programs by adding the military action threat tier that those programs do not address, using a methodology consistent with established dam safety engineering practice.

Section 3 — Facility Profile

3.1 Physical Systems

A large hydroelectric storage facility consists of several interacting physical systems whose functional relationships define both the generating-consequence chain and the dam-safety consequence chain. The following system descriptions apply to conventional concrete gravity, arch, or embankment dam configurations with surface or underground powerhouses. Facility-specific deviations are addressed in site-specific annexes when developed.

The impoundment structure (dam) retains the storage reservoir and provides the hydraulic head that drives generation. For concrete gravity and arch dams, the primary failure modes under military action are direct structural damage to the dam body and disruption of the foundation drainage and monitoring systems. For embankment dams (earth-fill or rock-fill), the primary failure modes are overtopping due to loss of spillway control, internal erosion accelerated by vibrations and disturbances caused by nearby detonations, and direct damage to the upstream slope protection.

The water conveyance system — intake structure, penstocks, and scroll casings — transports water from the reservoir to the turbine-generators. Penstock failure is the governing structural event for the generating consequence chain: a penstock rupture under full operating head produces a high-energy water jet that can cause catastrophic damage to the powerhouse and adjacent structures. The combination of a penstock failure and an upstream intake gate failure (stuck open) removes the operator's ability to control water discharge, potentially forcing the emergency spillway to operate.

The turbine-generator units convert hydraulic energy to electrical energy. Each unit consists of a water turbine (Francis, Kaplan, or Pelton configuration, depending on head and flow characteristics), a synchronous generator, and associated mechanical and electrical auxiliary systems. The unit step-up transformer (UST) steps the generator terminal voltage to transmission voltage for connection to the switchyard. USTs at large hydro facilities are typically single-phase or three-phase custom units rated for the specific voltage and MVA requirements of the generating unit; they are not interchangeable across facilities, and their replacement constitutes the governing long-lead-time equipment scenario.

The switchyard connects the generating units to the transmission system and serves as the point of interconnection where the facility's generating capacity enters the bulk electric system. The switchyard contains the high-voltage circuit breakers, disconnect switches, bus structures, and protection systems that govern the facility's connection to and disconnection from the transmission network. Under military action conditions, the switchyard is a high-value target because its damage simultaneously removes the facility from service and, depending on the transmission topology, may affect the wider transmission network.

The spillway and gate systems provide controlled release of reservoir water for flood management and reservoir level control. Spillway gates (radial, vertical-lift, or drum-gate configurations) are typically operated by electric drives with manual backup. Under military

action conditions, loss of spillway control — through power loss, control system disruption, or physical damage — removes the operator's primary tool for managing reservoir level in response to inflow events. This creates a condition in which dam safety consequences can escalate independently of the generating chain of consequences.

The dam safety monitoring system provides continuous measurement of the physical state of the dam structure: piezometric levels in the foundation and embankment, seepage rates and turbidity, settlement and movement survey benchmarks, and seismic accelerographs. These measurements are the primary basis for detecting developing dam safety conditions. Under military action conditions, the disruption of dam safety monitoring creates operational blindness to a potentially developing dam safety emergency. At the same time, the physical stresses on the structure may be elevated by the same events causing the operational disruption.

3.2 Grid Role and Synchronous Inertia Contribution

Large hydroelectric storage facilities perform functions in the electric grid beyond simple generation of electrical energy. Their specific grid contributions determine the consequence of their loss or degradation beyond the MW loss itself.

Synchronous inertia contribution is the most consequential grid function that distinguishes large hydro from other generation types at risk. Synchronous generators — rotating machines whose magnetic fields are locked to the grid frequency — contribute kinetic energy to the interconnection proportional to their moment of inertia and the square of the rotational speed. When generation is suddenly lost or load suddenly increases, this kinetic energy is released as the rotor decelerates, slowing the rate of frequency decline and providing time for governors and automatic generation control to respond. As non-synchronous renewable resources displace conventional synchronous generation, the interconnection's total synchronous inertia declines. Large hydro facilities represent a concentrated and irreplaceable source of synchronous inertia, particularly in the Western Interconnection, where WECC has documented declining inertia levels as a grid reliability concern.

The loss of a large hydro facility under military action conditions, therefore, produces two simultaneous frequency-related consequences: an immediate MW deficit that drives a frequency decline, and a reduction in total interconnection inertia that increases the rate of frequency decline for all subsequent events. The second consequence is persistent as long as the facility remains offline and compounds the impact on grid reliability of extended outages.

Voltage support and reactive power capability are the second grid function specific to synchronous generators. Large hydro units can be operated as synchronous condensers — spinning without generating real power — to provide reactive power support to the transmission system. Under military action conditions that damage the damage-generating capability but leave the machine intact, this condensing capability may be preserved even when the generating capability is lost. The design basis must account for the possibility that partial preservation of the

machine allows voltage support to continue even when the generating consequence chain is fully active.

Black start capability is held by a subset of large hydro facilities and designated in the CntlCo's operating plan as a grid restoration resource following a complete system blackout. Hydro facilities are well-suited for black start because they can start without an external power source — governor and excitation systems can be energized from station batteries, and the unit can be brought to rated voltage and frequency to energize the transmission system and pick up load sequentially. Under military action conditions that target multiple grid resources simultaneously, preserving black-start capability at large hydro facilities may be the critical path to grid restoration. Its loss compounds all other MA consequences.

3.3 Operational Characteristics Under Normal and Emergency Conditions

Large hydroelectric storage facilities operate under FERC-licensed conditions that govern minimum flow releases, seasonal reservoir-level targets, and operational constraints for downstream recreation, navigation, fish passage, and water supply. These license conditions create operational constraints that may limit the operator's ability to respond to emergency conditions: minimum flow obligations may prevent reservoir drawdown below a level that would otherwise improve the dam's safety margin; seasonal level targets may prevent reservoir modifications that would reduce hydraulic head on a damaged penstock.

Emergency operations are governed by the facility's Emergency Action Plan (EAP), which establishes notification procedures, condition levels, and downstream warning protocols for developing dam safety emergencies. The EAP is a required FERC Part 12 document and is tested periodically through tabletop and notification exercises. Under military action conditions, the EAP's notification procedures may be compromised by the same communications disruption that affects generating operations; its downstream warning systems may be damaged by the same attack that initiates the dam safety emergency; and the wider disruption of the military action event may impair its coordination with local emergency management agencies.

Section 4 — Design Basis: Postulated Events

4.1 Introduction

The four postulated event categories established in DBA-ES Section 3.4 are instantiated here for large hydroelectric storage facilities. Each category is described with its hydro-specific events, and the governing case within each category is identified. The governing case is the event configuration that produces the highest consequence at each tier of the consequence taxonomy and against which the acceptance criteria in Section 9 are assessed.

For large hydro facilities, the most consequential postulated event configurations involve simultaneous activation of multiple consequence chains. Events that simultaneously challenge

the generating function and the dam safety function — the dual-chain events — are governed cases for the facility class as a whole, not merely for an individual postulated event category. Section 5 develops the non-adversarial bounding events for each category; Section 6 applies the military action extension to produce the governing MA scenarios.

Code	Category	Hydro-Specific Events	Governing Condition
PE	Power Supply and Grid Interface	Loss of station service power; loss of auxiliary transformer; loss of governor power supply; failure of unit step-up transformer; loss of switchyard bus	Loss of unit step-up transformer (UST) — long-lead-time equipment analog to LPT at TransCo substations; replacement timeline 18–30 months for custom EHV units
SE	Structural and Physical Integrity	Penstock rupture or deformation; wicket gate failure (stuck open or closed); spiral casing damage; draft tube damage; powerhouse flooding; spillway gate failure; dam embankment seepage or slope instability; intake structure damage	Penstock rupture combined with spillway gate failure — simultaneous loss of controlled flow path and emergency discharge capacity; governs the downstream flood consequence chain.
CE	Control and Communications	Governor system failure or erroneous actuation; excitation system failure; SCADA loss at powerhouse; loss of protective relay coordination; failure of dam safety monitoring systems (piezometers, seepage sensors, settlement monitors); loss of communications to CntlCo	Governor system failure combined with loss of CntlCo communications — unit cannot respond to dispatch signals, and CntlCo cannot observe unit status; governs frequency response failure during grid event
XC	Cross-Sector Cascade	Gas supply curtailment reducing grid backup generation (gas-electric cascade); loss of water supply to downstream municipal systems; loss of irrigation supply affecting the agricultural sector; loss of navigation on regulated rivers; military or emergency service loss of black-start resources.	Simultaneous loss of GenCo generation and downstream water supply during extended outage — both consequence chains are active concurrently; no single sector-level mitigation addresses both

4.2 Dual-Chain Events

Several postulated events in the SE category activate both the generating and dam safety consequence chains simultaneously. These dual-chain events are identified separately because their analysis cannot be conducted as the superposition of two independent single-chain analyses. The physical interactions between the chains — the penstock failure that triggers both powerhouse damage and uncontrolled reservoir outflow, or the spillway gate failure that simultaneously reduces generation head and removes flood management capability — create a combined consequence that exceeds the sum of the individual chain analyses.

The primary dual-chain events for large hydro facilities are: penstock rupture combined with intake gate failure (loss of controlled conveyance with loss of shutdown capability); spillway gate failure during high inflow period (loss of flood management capacity concurrent with potential generation disruption); and embankment seepage acceleration following powerhouse

flooding (structural stress on the dam from reservoir water entering the powerhouse foundation). Each of these events is analyzed in Section 5 for the non-adversarial case and in Section 6 for the military action case.

Section 5 — Design Basis Accidents: Non-Adversarial Bounding Events

5.1 Governing Equipment Failure Events

The governing equipment failure event for the generating consequence chain is loss of the unit step-up transformer (UST) at maximum generation output. UST failure at full load results in an abrupt loss of generating capacity equal to the unit's full nameplate rating, without the controlled ramp-down that operational curtailment provides. For a facility with multiple units on a common bus connected through a single high-side breaker, the loss of the bus-section transformer is the governing single-contingency event; it removes multiple units simultaneously.

The recovery timeline for UST failure governs the extended-consequence analysis. A UST failure at the Grand Coulee scale — a single-phase 500 kV unit rated at several hundred MVA — has no compatible spare in the current STEP program inventory for most configurations. New procurement requires 18 to 30 months from order to energization. The non-adversarial DBA must bound this extended-consequence scenario: which grid reliability posture, which cross-sector consequence, and which downstream water management obligation persist during an 18-month generation outage at the facility?

The governing equipment failure event for the dam safety consequence chain is the failure of the primary spillway gate during periods of high reservoir inflow. Spillway gate failure — stuck closed, not open — at the onset of a large flood event removes the facility's primary flood management tool at the moment of peak demand. The consequence escalation path runs through reservoir overtopping (for embankment dams) or through stress on the dam structure that exceeds design-basis flood levels (for concrete dams). Gate failure stuck open is operationally disruptive but not a dam safety emergency; gate failure stuck closed during high inflow is the governing case.

5.2 Natural Hazard Bounding Events

The governing natural hazard events for large hydroelectric storage facilities are: probable maximum flood (PMF), which defines the maximum inflow the spillway system must safely pass; maximum credible earthquake (MCE), which defines the structural loads the dam and powerhouse must withstand; and extreme wind event causing wave action on the reservoir that exceeds the freeboard design allowance. FERC dam safety regulations require that facilities demonstrate their ability to pass the PMF and survive the MCE safely. These regulatory requirements define the baseline natural hazard design basis.

The compound-event scenario required by DBA-ES Section 4.3 for large hydro is a seismic event followed by a flood event within the recovery window — the dam sustains MCE-level structural stress, then faces an inflow event before the post-earthquake safety inspection is complete and repairs have been assessed. This compound event does not require that both events be at their respective governing magnitudes simultaneously; it requires that a moderate earthquake (above the operating basis earthquake but below the MCE) be followed by a significant inflow event before the facility is cleared for continued operation. The recovery window for post-earthquake inspection at a large dam is typically 30 to 90 days, which defines the compound-event temporal window.

5.3 Operational Bounding Events

The governing operational event for the generating chain is turbine runaway following load rejection with concurrent governor failure. Load rejection — the sudden disconnection of the unit from the grid by protective relay operation — causes the generator to accelerate rapidly as it transitions from motoring mode to spinning freely without load. The governor is designed to close the wicket gates immediately, limiting the rise in speed to below the mechanical runaway speed. If the governor fails to respond to the load rejection, the unit accelerates to runaway speed, which for large Francis turbines is approximately 150 to 200 percent of rated speed. Sustained operation at runaway speed places mechanical stress on the turbine runner, generator rotor, and bearings, which can cause catastrophic mechanical failure.

The governing operational event for the dam safety chain is an erroneous gate-opening command that opens the spillway gates when the reservoir level is already near the minimum operational level required for downstream flow obligations. While this does not directly threaten the dam structure, it represents a category of operational error — automated control system erroneous actuation — that, in different configurations (erroneous gate closure, erroneous power gate opening), has direct dam safety consequences. The DBA must bound the full range of erroneous actuation scenarios for all gate and valve systems.

5.4 Cross-Sector Cascade Bounding Events

The governing cross-sector cascade event for large hydro is the gas-electric cascade combined with extended generation outages. As described in DBA-ES Section 4.5, the gas-electric cascade creates a bidirectional dependency failure that neither the gas sector nor the electric sector can resolve independently. For large hydro, the cascade initiates when an extended generation outage at the hydro facility drives increased dispatch of gas-fired generation, which increases gas consumption and stresses pipeline transmission, potentially reducing gas availability for heating during peak demand periods. The addition of the downstream water-supply consequence chain — loss of minimum-flow releases affecting municipal water intakes downstream of the dam — creates a three-sector cascade (electric, gas, water) that has not been addressed in existing cross-sector planning frameworks.

Section 6 — Military Action Extension

6.1 Application of the Four Characteristics

The DBA-MA extension applies the four military action characteristics established in DBA-ES Section 5.2 to each postulated event category for large hydroelectric storage facilities. This section develops the governing military action scenario for each postulated event category. It identifies the specific pathways by which the four characteristics transform non-adversarial bounding events into their military-action analogs.

6.1.1 *Intentionality Applied to Large Hydro*

An adversary targeting large hydroelectric storage facilities chooses its targets from among the class based on strategic consequence. The strategic hierarchy within the facility class favors: facilities with black start designation (loss removes a grid restoration resource); facilities in high-inertia corridors of the Western Interconnection (loss compounds the existing inertia deficit); facilities whose loss produces simultaneous downstream flood consequence (dual-chain events achieve compound strategic effect per weapon); and facilities with long lead times for critical equipment where no comparable spare exists.

At the facility level, intentionality directs targeting toward the components whose loss produces the highest consequence per unit of effort: the unit step-up transformers (which are outdoor, minimally protected, and irreplaceable on short timelines); the spillway gate control systems (which can be disrupted to remove flood management capability and accelerate dam safety consequence escalation); and the dam safety monitoring systems (which, if disrupted, create operational blindness to developing dam safety conditions). The adversary does not need to destroy the dam to achieve strategic effect — it needs to remove the operator's ability to manage the impoundment safely.

6.1.2 *Persistence Applied to Large Hydro*

The persistence characteristic is particularly consequential for large hydro, given the replacement timelines for critical equipment. A UST destroyed by a precision strike cannot be replaced from existing inventory on a timeline that prevents the extended-consequence regime. If the adversary can prevent replacement equipment from being manufactured, transported, or installed — through continued denial of site access, supply chain interdiction, or targeting of manufacturing facilities that produce large power transformers — the generation outage persists indefinitely. The non-adversarial DBA bounds the consequence of an 18-to-30-month UST replacement timeline. The MA extension must bound the consequence of an indefinitely extended outage where the replacement timeline itself is contested.

For the dam safety consequence chain, persistence operates through the denial of maintenance access. Dam safety depends on continuous monitoring, periodic inspections, and timely

maintenance of drainage systems, gate mechanisms, and structural monitoring instruments. Civilian crews cannot safely maintain a facility under military occupation or persistent threat of precision strike. The FERC Part 12 inspection schedule and the EAP notification requirements are both predicated on qualified personnel's ability to access the facility. Under persistent denial conditions, dam safety can degrade over weeks to months, even without a direct structural attack.

6.1.3 Precision Applied to Large Hydro

Precision enables targeted functional degradation beyond simple physical destruction. For the generating consequence chain, precision targeting of the unit step-up transformer achieves the maximum electrical consequence — permanent loss of generation — while limiting collateral damage to the turbine-generator and other mechanical components that would otherwise be usable for partial operation. Precision targeting of the governor and excitation control systems, rather than the turbine itself, results in turbine runaway during the next load-rejection event without visibly damaging the facility at the time of attack.

For the dam safety consequence chain, precision targeting of the spillway gate drive systems or the gate control house produces loss of spillway control without structural damage to the dam body. This creates a latent dam safety emergency: the dam structure appears undamaged, normal operations appear to continue, but the operator's ability to manage reservoir level in response to inflow events has been removed. Detection of this condition requires manual inspection of the gate drive systems, which may be denied access to. Precision thus creates the conditions for a dam safety emergency that develops on a timeline governed by hydrology — the next significant inflow event — rather than by the timing of the original attack.

6.1.4 Denial Applied to Large Hydro

The denial characteristic produces its most consequential effects at large hydro facilities through the interaction between the generating-consequence chain and the dam-safety consequence chain. Dam safety fundamentally depends on human access: piezometers must be read, seepage measured, gates manually tested, and trained personnel present to assess evolving conditions. Loss of site access for dam safety personnel removes the primary means of detecting and responding to developing dam safety emergencies.

Under denial conditions, the EAP notification chain fails at its origin — the site personnel responsible for observing and reporting adverse conditions are not present. Downstream warning systems that require site personnel to activate cannot be activated. The FERC notification obligation cannot be met because the required observations cannot be made. Denial thus transforms an impaired dam safety monitoring condition into complete operational blindness to the facility's dam safety state, at precisely the time when that state is most at risk.

6.2 Governing Military Action Scenarios

Two scenarios for governing military action are identified for large hydroelectric storage facilities. Each governs a different consequence chain; both must be addressed in the MA extension analysis.

Governing MA Scenario 1 — Precision Strike on UST with Persistent Denial: A precision strike destroys the unit step-up transformer at the highest-rated generating unit. Site access is denied for equipment assessment and emergency repair. A compatible spare is not available from STEP inventory. Generation from the unit is lost for the duration of the procurement-and-installation timeline, subject to persistent interdiction. This scenario governs the extended-consequence analysis for the generating chain: it produces the longest generation outage, the greatest synchronous inertia loss, the longest degradation of potential black start capability, and the longest cross-sector cascade exposure.

Governing MA Scenario 2 — Precision Strike on Spillway Control with Persistent Denial During High-Inflow Period: A precision strike destroys the spillway gate drive systems or the gate control building. Site access is denied for repair. A significant inflow event (at or above the 100-year flood level, below the PMF) occurs during the denial period. The operator cannot open the spillway gates to pass the inflow. Reservoir level rises toward overtopping. This scenario governs the dam safety consequence chain analysis: it produces the highest probability of structural dam safety consequence under MA conditions and activates the downstream flood wave consequence chain.

These two governing scenarios are not mutually exclusive. A coordinated attack could initiate both simultaneously. The dual-chain scenario — simultaneous precision strikes on the UST and the spillway control, followed by persistent access denial during a high-inflow period — is the compound-governing case that yields the highest combined consequence per facility.

Section 7 — Consequence Taxonomy: Dual Chain Analysis

7.1 Overview of the Dual Chain Structure

Large hydroelectric storage facilities present a consequence structure that extends the DBA-ES three-tier taxonomy in one critical respect: both the generating consequence chain and the dam safety consequence chain must be analyzed at each tier, and the combined consequence of simultaneous activation of both chains must be assessed separately because the combined consequence exceeds the sum of the individual analyses.

The generating consequence chain propagates through the electric grid: Tier 1 is the facility-level generation loss; Tier 2 is the grid-level reliability consequence of that loss (frequency, voltage, stability, inertia); Tier 3 is the cross-sector consequence of the grid reliability degradation. This is the standard DBA-ES consequence taxonomy applied to a GenCo asset.

The dam safety consequence chain propagates through the physical geography of the watershed: Tier 1 is the structural condition of the dam (impaired but intact, or failing); Tier 2 is the downstream flood consequence within the immediate watershed (inundation of the flood zone below the dam); Tier 3 is the cross-sector consequence of the flood (loss of downstream water supply infrastructure, transportation corridor loss, municipal inundation). The dam safety Tier 3 reaches cross-sector cascading consequences through a different physical pathway than the electric grid, but the consequence taxonomy structure is the same.

7.2 Generating Consequence Chain

Tier 1 — Facility-level generating consequence. The immediate consequence of the governing MA Scenario 1 is the loss of generating capacity equal to the affected unit's nameplate rating, instantaneously disconnected from the grid without the ramp-down that distinguishes controlled curtailment from a fault-induced trip. For a 500 MW unit, this is a 500 MW step change in generation available to the interconnection. For a multi-unit facility where the governing event affects a common bus, the step change may be proportionally larger.

Tier 2 — Grid reliability consequence. The grid-level consequence is assessed under BAL-003 using the interconnection's documented frequency response for a generation loss of the magnitude produced by the governing event. For Western Interconnection facilities at Grand Coulee or John Day scale, the generation loss from a full facility trip may approach or exceed the largest event the interconnection's frequency response obligation is sized to address. For Eastern Interconnection facilities, the relative magnitude is smaller, but the consequences for regional reliability may still be significant.

The synchronous inertia consequence is a second Tier 2 effect that persists for the duration of the outage, independent of the immediate frequency response. As long as the facility remains offline, the interconnection operates with less total inertia. The rate-of-change-of-frequency (ROCOF) for all subsequent generation loss events is higher than it would be with the service facility. This persistent inertia deficit is a background condition that increases the severity of all other grid reliability events occurring during the extended outage period.

Tier 3 — Cross-sector cascade from the generating chain. The extended generation outage drives increased dispatch of gas-fired generation, increasing gas consumption on interstate pipelines. If the facility carries a black start designation and that designation is lost, the CntlCo's restoration plan following a subsequent blackout event is degraded. Military or government loads directly served from the facility's switchyard transition to backup generation, consuming backup fuel reserves on a timeline governed by the outage duration. Water management for downstream beneficial uses shifts from turbine discharge to spillway release or reservoir drawdown, depending on the spillway system's operational state.

7.3 Dam Safety Consequence Chain

Tier 1 — Facility-level dam safety consequence. The immediate consequence of the governing MA Scenario 2 is loss of spillway control — the operator cannot release water through the primary flood management system. Reservoir level begins to rise as inflow exceeds the turbine discharge capacity. If turbine units are also offline (dual-chain event), the only controlled-release path is through any auxiliary or emergency outlets capable of independent operation. The Tier 1 condition is: impaired structural safety with rising reservoir level, denied access for assessment, and no EAP notification because the triggering observation cannot be made.

Tier 2 — Downstream watershed consequence. If the reservoir level reaches the overtopping threshold before the spillway control is restored or before the inflow subsides, uncontrolled flow over the dam crest begins. For embankment dams, overtopping erosion can initiate rapidly and lead to full dam failure within hours of overtopping onset. For concrete dams, overtopping may be tolerable for a limited period, depending on the dam geometry and flow rate, but a structural assessment requires site access that may be denied. The Tier 2 consequence is the downstream flood wave from an uncontrolled release or structural failure, which reaches the flood zone below the dam within a time-of-travel period ranging from minutes (steep-gradient canyons) to hours (broad agricultural valleys).

Tier 3 — Cross-sector cascade from the dam safety chain. The downstream flood wave inundates infrastructure in the flood zone, including municipal water intakes, wastewater treatment plants, transportation corridors, communication infrastructure, and any industrial or military facilities sited in the valley below the dam. The loss of downstream water supply infrastructure can persist for months to years following the flood event as water treatment and distribution systems are assessed and rebuilt. For rivers with multiple downstream dams, the flood wave from an upper dam failure may threaten lower dams with overtopping or foundation erosion.

7.4 Combined Consequence Assessment

When both consequence chains are activated simultaneously by a coordinated attack, the combined consequence assessment must address interactions between the chains that cannot be captured in independent analyses of each. The governing interactions are:

Operations personnel are divided between responding to the generating emergency and the dam safety emergency, with neither receiving full attention. The CntlCo is receiving loss-of-generation notifications and requesting status information while the facility operator is attempting to assess dam safety conditions and activate EAP notifications. The communications and personnel resources available to the facility operator are insufficient to manage emergencies effectively simultaneously.

The loss of turbine discharge — the primary low-level release path for reservoir drawdown — removes the operator's ability to reduce reservoir level ahead of an inflow event. Under the non-adversarial DBA, turbine discharge is available as a reservoir management tool even during generating equipment outages if the turbine wicket gates can be operated independently of the

generator. Under MA conditions where the generating units have been taken out of service by attack, the continued availability of turbine discharge requires that the wicket gate hydraulic systems be intact and operable. This cannot be assumed following a precision strike on the UST or the generator.

The extended consequence duration of the combined scenario is governed by the longer of the two recovery timelines: the UST replacement timeline (18 to 30 months) for the generating chain, or the dam repair and safety recertification timeline (18 to 30 months) for the dam safety chain if structural damage has occurred. If structural damage to the dam body requires repair before the reservoir can be refilled to normal operating level, the facility's generating capacity may be unavailable for the duration of dam repair — potentially extending beyond the UST replacement timeline.

Section 8 — Protective Action Principles

8.1 Defense-in-Depth Architecture

The protective action architecture for large hydroelectric storage facilities under both non-adversarial and military action conditions applies the defense-in-depth logic established in DBA-ES Section 7.1. Protective measures are layered from passive physical hardening through active operational response to external coordination, with the local sovereignty principle requiring that critical safety functions remain sustainable without external power, communications, or operator action for the design-basis duration.

For large hydro facilities, the critical safety functions requiring passive protection are: controlled impoundment (the ability to manage reservoir level without active intervention), turbine unit shutdown (the ability to stop generating units safely without external power or communications), and dam safety monitoring (the ability to observe and report the structural state of the dam without external power or personnel access). Each of these functions must be analyzed for its passive protection capability, and the results must be expressed as a design basis duration: how long can the facility maintain safe conditions without active intervention?

8.2 Physical Hardening Priorities

The physical hardening priority hierarchy for large hydro facilities is established by the combination of consequence severity and replacement lead time. The three highest-priority physical hardening targets are:

Unit step-up transformers are the highest priority. Their replacement lead time (18 to 30 months for custom EHV units) is the governing parameter of the extended-consequence scenario. Physical hardening measures include: protective berms or blast-resistant enclosures around outdoor transformer installations; relocation of oil-filled transformer banks away from facility

perimeters where practical; and installation of fire suppression systems capable of limiting fire damage to transformer windings and bushings in the event of blast damage.

Spillway gate drive systems and control buildings are the second priority. The loss of spillway control triggers the dam safety consequence chain. Hardening measures include: redundant gate drive systems (hydraulic and electric with independent power supplies); hardened or underground gate control buildings; and installation of manual override mechanisms that allow gate operation without power or control systems.

Dam safety monitoring infrastructure is the third priority. Passive monitoring instruments (vibrating-wire piezometers, settlement benchmarks) that do not require power to operate represent the most survivable monitoring configuration. Battery-backed remote data loggers provide a power-independent data collection point. Hardened communications paths (satellite uplinks, HF radio) ensure that monitoring data reaches dam safety authorities even when primary communications are disrupted.

8.3 Bright Line and Command Broker Requirements

The Bright Line, defined in FD-BL v0.2 (§4.2–§4.4) and applied to this facility class by reference to DBA-ES-SF, applies to the governor, excitation control, and gate control systems of large hydro facilities. These systems directly control physical processes — the turbine wicket gates, the generator excitation, and the spillway gate positions — whose erroneous actuation can affect the dam safety chain as well as the generating chain. An action that commands a change to any of these physical process parameters is above the Bright Line when its incorrect performance could reach the facility's severity threshold — including the dam-safety consequence chain — and requires human Command Broker authorization enforced architecturally (§4.5), whatever component originates it. Determinism governs only the scope carve-out (§4.4): a component deterministic within a stated bound and formally testable as deployed may perform monitoring and alarming actions below the line autonomously, but a deterministic component can still originate an above-line command.

The command-broker requirement for gate control systems is particularly important under military action conditions because automated gate control commands — received from a compromised control system or injected through a cyber intrusion coincident with a physical attack — could open or close spillway gates at operationally dangerous times. The command broker architecture must ensure that gate-position commands, regardless of their origin within the control system, are confirmed by a qualified human operator before execution. For remote facilities where operator presence may be reduced under MA conditions, this requirement must be addressed by the facility's emergency operating procedures.

8.4 CntlCo Coordination and Black Start Sequencing

The recovery sequencing governing principle from DBA-ES Section 7.3 applies to large hydro facilities, with an additional dimension: facilities with black-start obligations have a role in the CntlCo’s system restoration plan that takes priority over their own generation restoration sequence. Under MA conditions in which a grid blackout has occurred, the CntlCo may request that a large hydro facility activate its black-start capability before its own generating damage has been fully assessed and repaired. The facility’s emergency operating procedures must address this scenario: under what conditions can and should the facility attempt black-start operations while in an impaired generating or dam-safety state?

The answer to this question requires coordination between the facility operator, the FERC-licensed dam safety program, and the CntlCo before the MA event — it cannot be resolved in real time during a grid emergency. The design basis must include a documented agreement between these parties establishing the conditions under which black start activation is appropriate despite facility impairment, and the conditions under which dam safety concerns override grid restoration priority.

Section 9 — Acceptance Criteria Framework

9.1 Framework Structure

Acceptance criteria for large hydroelectric storage facilities are expressed at each tier of the dual-chain consequence taxonomy and for both the non-adversarial DBA and the MA extension. The criteria are organized into three types — physical, operational, and cross-sector — consistent with the three required adaptations for electric sector design-basis application established in DBA-ES Section 3.2.

The dam safety acceptance criteria constitute a fourth category specific to this facility class. Dam safety criteria are defined by reference to FERC Part 12 regulatory thresholds and USCOLD guidelines, expressed in terms of the physical parameters monitored by the dam safety surveillance and monitoring program. These criteria apply to the dam safety consequence chain at Tier 1 and govern the conditions that trigger EAP activation.

Tier	Criterion Type	Non-Adversarial DBA Standard	MA Extension Standard
Tier 1	Physical	No structural damage to the dam embankment, intake, penstock, or powerhouse that cannot be remediated within the facility’s maintenance window without triggering dam safety notification obligations under FERC Part 12.	Facility maintains safe shutdown of all generating units and controlled impoundment without on-site personnel access for a minimum of 72 hours following the initiating event.
Tier 1	Operational	Generating capacity restored to at least 50 percent of rated output within 30 days of initiating event; full capacity restored within 90 days except for long-lead-time	Generating capacity status is accurately reported to CntlCo within 4 hours of initiating the event via backup communications path independent of the

		equipment failures.	primary SCADA link.
Tier 2	Grid reliability	A facility trip does not, by itself, produce a frequency excursion exceeding NERC BAL-003 limits or trigger automatic under-frequency load shedding within the interconnection.	If the facility carries a black start designation, backup start capability (diesel or battery) survives the initiating event and is available to CntlCo within 6 hours of request.
Tier 3 — cascade	Cross-sector	An extended outage (>30 days) does not reduce downstream water availability below minimum flow obligations under FERC license or applicable water rights agreements.	Downstream municipal water supply systems, irrigation systems, and navigation channels have backup supply or storage sufficient for 14 days independent of facility operation.
Tier 3 — direct dependency	Cross-sector	Any military, government, or defense-critical load directly supplied from the facility's switchyard has backup generation sufficient for 72-hour independent operation.	Direct dependency loads are identified and reported to FERC and relevant defense agencies; their backup posture is validated at each FERC Part 12 periodic inspection.
Dam safety	Structural/legal	No condition arises that requires Emergency Action Plan (EAP) notification under the FERC dam safety program. Seepage, settlement, and piezometric levels remain within Action Level 1 thresholds.	EAP notification and downstream warning system remain operational under all MA conditions, including loss of primary communications. Backup notification path (radio, satellite) is tested quarterly.

9.2 Acceptance Criteria for Extended Outage Scenarios

The extended outage scenario — generation loss persisting beyond 90 days due to long-lead-time equipment replacement — requires acceptance criteria that address the background grid reliability posture during the outage period, not only the post-restoration condition. These extended-outage acceptance criteria are:

- The CntlCo must be able to demonstrate, within 30 days of the initiating event, that the interconnection can meet its BAL-003 frequency response obligation without the affected facility's inertia contribution. If it cannot, compensating measures (inertia-equivalent synthetic resources or revised under-frequency load-shedding settings) must be implemented within 60 days.
- For facilities carrying a black start designation, the CntlCo must identify and activate an alternative black start resource within 30 days of the initiating event. The alternative resource must be exercised within 60 days to confirm operability.
- Downstream water management — minimum flow releases for municipal supply, irrigation, and environmental obligations — must be maintained throughout the extended outage period through spillway operation or alternative release mechanisms. If no alternative release mechanism is available without site access, the acceptance criterion cannot be met, and notification to the dam safety authority is required.

Section 10 — Gap Analysis Framework

10.1 Purpose and Application

The gap analysis framework identifies a structured method for comparing a specific facility’s current capability against the design-basis requirements established in Sections 4 through 9. The gap analysis produces a set of findings — conditions in which current capability falls below design-basis requirements — that support investment prioritization and the demonstration of regulatory compliance.

For large hydro facilities, the gap analysis is conducted separately for the generating-consequence chain and the dam-safety consequence chain, with a combined assessment that identifies gaps whose closure addresses both chains simultaneously. Dual-chain gap closure — a hardening measure that improves both generation and dam safety resilience — is the highest-investment-efficiency category and should be prioritized accordingly.

For federally owned and operated facilities, the gap analysis findings translate into capital project requests within the agency’s budget process rather than into FERC license compliance obligations. The findings structure is the same; the enforcement pathway differs.

Gap Category	Gap Description	Assessment Method	Investment Priority Logic
Physical hardening	Inadequate physical protection of the powerhouse, switchyard, and intake structure against direct attack or precision strike.	Compare current physical security measures against CIP-014 R3 security plan requirements; identify unprotected critical components.	Priority proportional to the consequence tier of the unprotected component and the replacement lead time. UST hardening is the highest priority given the lead time.
Long-lead-time equipment	Absence of pre-positioned spare for unit step-up transformer or turbine-generator components with >12-month procurement lead time.	Inventory critical equipment; compare against STEP program coverage and facility-specific spare inventory. Identify gaps by voltage class and MVA rating.	Priority is highest where the spare gap coincides with a single-point-of-failure topology. Joint procurement through STEP or a regional compact reduces unit cost.
Control system independence	Governor, excitation, and gate control systems share a common power supply, communications infrastructure, or software platform that constitutes a single point of failure under MA conditions.	Review control system architecture for shared dependencies; test manual backup operation capability for each critical control function.	Priority is highest where shared dependency affects both the generating function and the dam safety monitoring simultaneously.
Communications independence	Absence of backup communications path to CntlCo and dam safety authority independent of primary fiber or microwave link.	Test backup communications (satellite, HF radio) under simulated primary path loss; verify CntlCo can receive unit status and dispatch commands	Low capital cost relative to the consequence of CntlCo communications loss during the MA event. Priority is high for facilities in an identified cluster

		via backup.	or corridor.
Black start capability	Degradation or loss of black start capability due to inadequate maintenance of diesel or battery start systems, or absence of backup fuel supply.	Test the black start sequence under simulated loss of all external power; verify diesel fuel inventory and the delivery contract for an extended outage scenario.	Priority is highest for facilities that carry a formal black-start obligation in the CntlCo operating plan. Loss of black-start capability removes a grid-restoration resource at the moment of highest demand.
Dam safety monitoring	Dam safety monitoring systems (piezometers, seepage sensors, settlement monitors) depend on AC power, with no battery backup or manual observation capability.	Identify monitoring instruments that have no battery or independent power backup; assess manual observation capability and frequency in a loss-of-power scenario.	Priority is highest where the monitoring gap affects Action Level 1 threshold parameters. FERC Part 12 periodic inspection provides a regulatory enforcement point.

10.2 Gap Closure Prioritization Sequence

Gap closure investments are prioritized in the following sequence, applied after all gaps are identified:

Priority: gaps whose open condition creates an acute dam safety risk. Any gap in dam safety monitoring, spillway control, or EAP notification capability that leaves the facility unable to detect and respond to a developing dam safety emergency takes precedence over all other investment categories. Dam safety acute risk gaps must be closed within one FERC Part 12 inspection cycle.

Second priority: long-lead-time equipment gaps where no compatible spare exists. Given that procurement lead times of 18 to 30 months cannot be compressed, the decision to procure a spare UST or turbine-generator component must be made years before the gap has consequences. Facilities with no compatible spare in inventory or accessible through a regional compact are in a structural gap condition that worsens over time in the absence of active procurement.

Third priority: control system independence and communications gaps. These gaps typically have lower capital costs than equipment procurement and can be closed within a normal capital maintenance cycle. The consequence of the gap determines their relative priority: a communications gap that prevents CntlCo coordination is higher priority than one that only affects internal monitoring.

Fourth priority: physical hardening improvements beyond the CIP-014 security plan baseline. Physical hardening improvements not required by the CIP-014 security plan but warranted by the MA extension analysis constitute discretionary capital investment. Their priority within the fourth category is governed by the target's consequence tier when hardened.

Section 11 — Deferred Items and Future Development

11.1 Deferred Items

The following items are identified for resolution in subsequent revision passes of this document.

DI-1: Quantitative inertia consequence analysis. Section 7.2 identifies the synchronous inertia consequence of a large hydro outage as a persistent Tier 2 effect but does not quantify it. Quantitative assessment requires facility-specific inertia constants (H constants) and the applicable interconnection's current inertia baseline. Resolution criterion: H constant data and WECC/NERC inertia baseline incorporated at next revision; quantitative ROCOF impact expressed for the governing case.

DI-2: Downstream flood consequence mapping. Section 7.3 qualitatively describes the dam safety Tier 2 and Tier 3 consequence chains. Meaningful acceptance criteria for the dam safety chain require facility-specific inundation mapping — the downstream flood zone, the infrastructure within it, and the flood wave's time of travel. Resolution criterion: FERC-required inundation maps (Exhibit H of FERC license) incorporated by reference at next revision; cross-sector dependency clients within the inundation zone identified.

DI-3: Pumped storage companion instrument. Pumped storage facilities are excluded from the scope in Section 1.3 because their consequence profile differs materially from that of conventional storage hydro. However, the growth of large-scale pumped storage as a grid resource means this exclusion increasingly represents an analytical gap. Resolution criterion: pumped storage addressed in a companion instrument or annex following completion of this instrument's first full review cycle.

DI-4: Federal facility gap analysis pathway. Section 10.1 notes that the gap closure enforcement pathway differs for federally owned facilities. The capital planning processes and budget cycle for USACE, USBR, and BPA facilities require a different framing of gap analysis findings than the FERC license compliance framing used for private licensees. Resolution criterion: federal facility gap analysis pathway section added at next revision following engagement with relevant agency planning offices.

DI-5: CntlCo black start agreement template. Section 8.4 requires a predocumented agreement among the facility operator, the dam safety authority, and CntlCo that establishes conditions for black-start activation during facility impairment. No template for such an agreement currently exists. Resolution criterion: draft agreement template developed in coordination with NERC and at least one Western Interconnection CntlCo; incorporated as an appendix at next revision.

11.2 Future Development

The following directions are anticipated for subsequent development of this document and the series.

Site-specific annexes for representative reference facilities will be developed as the series engages with GenCo operators. Grand Coulee, Hoover, and Robert Moses Niagara are the anticipated first-tier annex facilities, given their strategic significance and the completeness of the public-domain dam safety and reliability data available for them.

An international annex for allied-nation hydroelectric facilities is anticipated as the DBA-ES series develops its international engagement. The Kakhovka precedent demonstrates that large hydroelectric facilities are confirmed military targets in peer-adversary conflict. European facilities — particularly in NATO member states with significant hydro capacity and proximity to the eastern threat environment — are the anticipated first-tier international annex targets.

Integration with the PMU position paper (Roxey, Cummings, Dagle) is anticipated at the next revision. The FWHM wavefront analysis for predictive UFLS ride-through extension applies directly to the frequency response consequence chain analysis in Section 7.2. The governing MA scenario involves a large generation-loss event at a hydro facility during a period of reduced interconnection inertia — precisely the scenario the PMU position paper addresses. Cross-referencing once the position paper is complete will strengthen the quantitative basis for the Tier 2 frequency consequence analysis.

Section 12 — References

References are listed in the order of their first appearance in the document. A formal references section is a set of standards established in DBA-ES v0.4, effective for all new and revised instruments from that version onward.

Regulatory and Standards References

Federal Energy Regulatory Commission. 18 CFR Part 12: Safety of Water Power Projects and Project Works. Washington, D.C.: FERC. Current version.

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

North American Electric Reliability Corporation. FAC-001-3: Facility Interconnection Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. FAC-002-3: Facility Interconnection Studies. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. PRC-024-3: Generator Frequency and Voltage Protective Relay Settings. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. BAL-003-2: Frequency Response and Frequency Bias Setting. Atlanta: NERC. Current version.

United States Society on Dams (USSD, formerly USCOLD). Recommended Guidelines for the Safety Inspection of Dams. Denver: USSD. Current edition.

United States Society on Dams (USSD). Guidelines for Dam Safety Emergency Action Planning. Denver: USSD. Current edition.

Government and Policy References

Defense Science Board Task Force on DoD Energy Strategy. More Fight — Less Fuel. Washington, D.C.: Office of the Under Secretary of Defense for Acquisition, Technology, and Logistics, February 2008. Available at: <https://apps.dtic.mil/sti/pdfs/ADA477619.pdf>

Government Accountability Office. Defense Critical Infrastructure: Actions Needed to Improve the Identification and Management of Electrical Power Risks and Vulnerabilities to DoD Critical Assets. GAO-10-147. Washington, D.C.: GAO, October 2009.

U.S. Department of Energy and North American Electric Reliability Corporation. Spare Equipment Database (SED) and Spare Transformer Equipment Program (STEP). Washington, D.C.: DOE. Program documentation, current version.

Department of Homeland Security. National Infrastructure Protection Plan 2013: Partnering for Critical Infrastructure Security and Resilience. Washington, D.C.: DHS, 2013.

Evidentiary References

Nova Kakhovka Dam failure, Kherson Oblast, Ukraine, June 6, 2023. First confirmed deliberate destruction of a major hydroelectric storage facility in the modern threat environment. Primary consequence chains: loss of approximately 334 MW of GenCo generating capacity; downstream flood wave inundating Kherson Oblast; loss of Kakhovka Reservoir as a cooling-water source for the Zaporizhzhia Nuclear Power Plant. Open-source documentation: UNOSAT satellite analysis; Ukrainian government reporting; IAEA situation reports.

Series Parent and Cross-Reference Documents

Roxey, T. DBA-EN-SF: Energy Sector Framework. Eclectic Technologies, v0.8, 2026.

Roxey, T. DBA-ES-Sector_Framework: Electric Sector Design Basis Framework. Eclectic Technologies, v0.9, 2026.

Roxey, T. DBA-ES-MA-SF: Electric Sector Military Action Sector Framework. Eclectic Technologies, v0.2, 2026.

Roxey, T. DBA-ES-MA-QB: Electric Sector Military Action Quantitative Design Basis. Eclectic Technologies, v0.1, 2026.

Roxey, T. DBA-GAS-Sector_Framework: Natural Gas Sector Design Basis Framework. Eclectic Technologies, v1.2, 2026.

Roxey, T. DBA-DC-L1: Hyperscale Data Center Design Basis Framework. Eclectic Technologies, v1.0, 2026.

Roxey, T. DBA-MA Introduction. Eclectic Technologies, v1.7, 2026.

Revision History

Version	Date	Author	Description
0.1	May 2026	T. Roxey	Initial working draft. First facility-class instrument in the Electric Sector Design Basis Series following establishment of the DBA-ES sector parent (v0.2). Established a dual-consequence-chain analysis framework for large hydroelectric storage facilities. Instantiated all four sector-level postulated event categories for the hydro facility class. Applied DB → DBA → DBA-MA three-layer progression. Incorporated Kakhovka as the primary evidentiary anchor. Released for technical review.
0.2	May 2026	T. Roxey	Added Section 12 — References. Established formal references section per series standard set at DBA-ES v0.4. References cover FERC Part 12 dam safety program, NERC reliability standards (FAC-001/002, CIP-014, PRC-024, BAL-003), USSD dam safety guidelines, DSB “More Fight — Less Fuel” (2008), GAO-10-147, DOE/NERC STEP program, NIPP 2013, Kakhovka evidentiary anchor, and series parent/cross-reference documents.
0.3	May 2026	T. Roxey	Housekeeping: DBA-ES parent version reference corrected from v0.2 to v0.4 in the cover child-document line and Section 1 introduction. Revision history rows at v0.1 and v0.2 retain their original v0.2 references as accurate historical record of the parent version at those revisions. Addresses Gap 6.6 of DBA Series Gaps and Recommendations v1.2.
0.4	June 2026	T. Roxey	Housekeeping: DBA-ES parent version reference corrected from v0.4 to v0.6 in the cover child-document line, Section 1 introduction, and references section citation. Metadata parent document field updated from v0.2 to v0.6. Revision history rows at v0.1, v0.2, and v0.3 retain their original references as accurate historical record of the parent version at each prior revision. DBA-ES is now at v0.6; the lag from v0.4 to v0.6 has accumulated since the v0.3 correction pass.
0.5	June 2026	T. Roxey	N-15/N-16 batch pass (Tier B). Four changes: (1) Parent updated from DBA-ES v0.6 to v0.9; series grandparent DBA-EN-SF v0.8 added to cover and position block. (2) §1.4 MA Extension Applicability rewritten as four-filter structured assessment per DBA-

			ES-MA-SF v0.2 §1.4 and QB v0.1 §5 — all four filters satisfied at class level; FC1 identified as highest-consequence physical target in the DBA-ES series (dual-consequence chain; Kakhovka evidentiary anchor). Closes N-15/OI-8 for HY-FC1. (3) §1.4a MA Communications Design Basis added — CB-1 through CB-5 instantiated with FC1-specific dual backup channel obligation (CntlCo and dam safety authority) and CB-2 remote-siting pre-authorized local authority constraint for dam safety emergency actions. Closes N-16/OI-1 for HY-FC1. (4) DBA-EN-SF v0.8, DBA-ES-MA-SF v0.2, DBA-ES-MA-QB v0.1 added to references; DBA-ES updated to v0.9. Revision history moved to end of document.
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